

Zero-length Springs and Slinky Coils

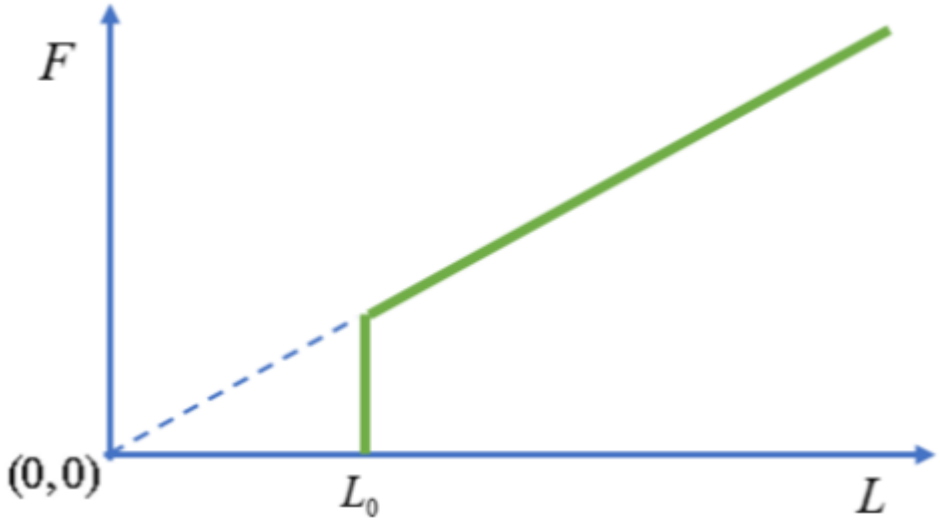
Theory Question 1

Presentor: Niv Cohen
Authors: Eli Raz, Itamar Hason

Part A, Zero-length spring

- Definition of Zero-length spring (ZLS):





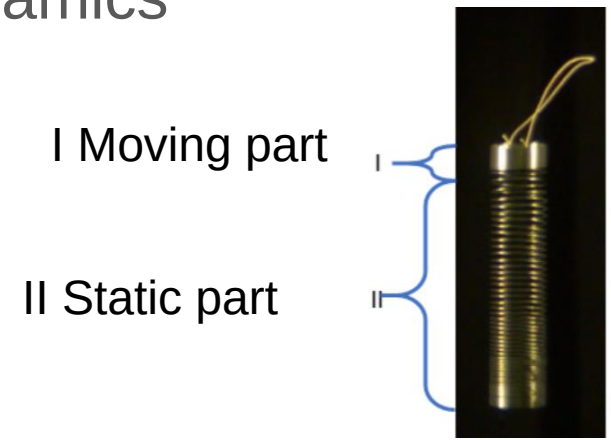
Part A, Statics -
Conclusions the the student should reach

- Every part of a Zero-Length Spring (ZLS) is a ZLS
- Small spring element has large spring constant



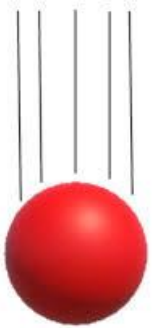
Part B, Dynamics

- The bottom part remain static until the moving part reaches it
- We ask the students to calculate properties of the fall given this kind of dynamics



Part B, Dynamics
Conclusions the the student should reach

- Center of mass position determines the fall rate of the spring





Part B, Dynamics

- The slinky decelerate along the fall (needs to accelerate larger mass densities per unit time)



Part B, Dynamics - tasks

- Calculate the total fall time:
  - Center of mass free falls from its initial stretched position to the collapsed one

- Calculate the velocity of the moving part

Part C, Energetics

- Energy is lost as a result of adding coils to the moving part







Q2:The Physics of a Microwave Oven

Gal Dor, Ofer Eyal, Moshe Goldstein, Eli Raz

Goals

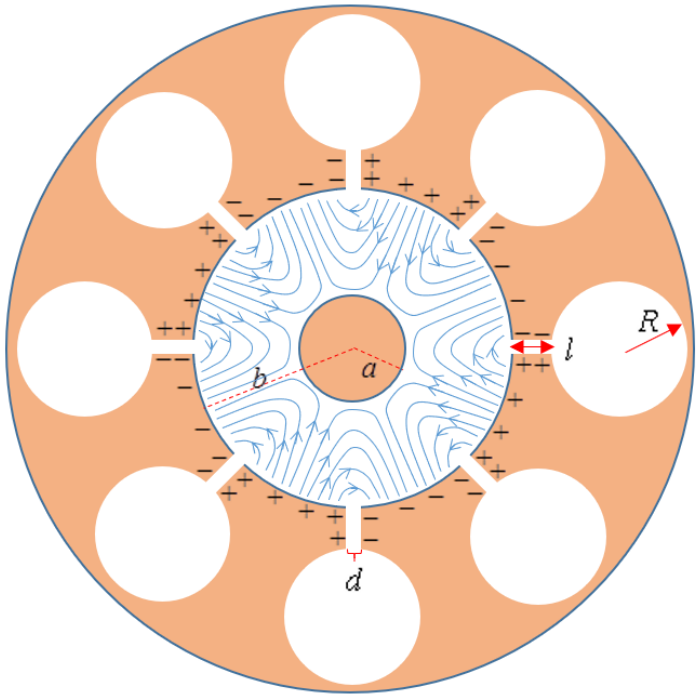
- Part A: Principles of magnetron operation
- Part B: Dielectric absorption of microwave radiation by water

- This talk: very quickly go over key ideas
  - … Apologies in advance for quality of animations.

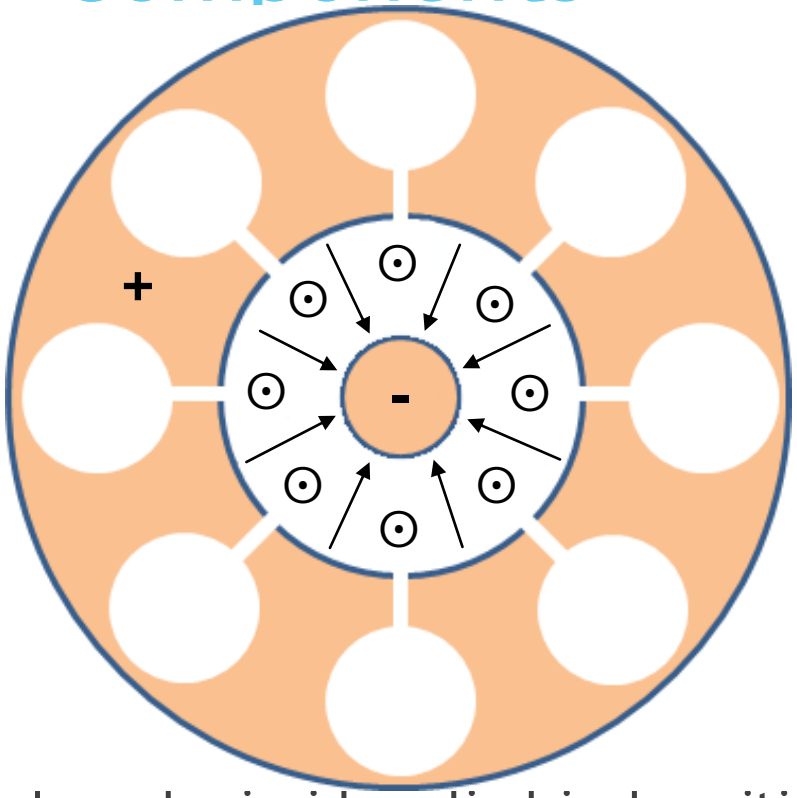
Part A: Magnetron

Magnetron

- Magnetrons are complex, non-linear systems.
- Aim: qualitative understanding of the principles of magnetron operation.
  - Electron drift in electric and magnetic fields
  - Focusing of electrons by oscillating cavities
- Short discussion of magnetron operation.



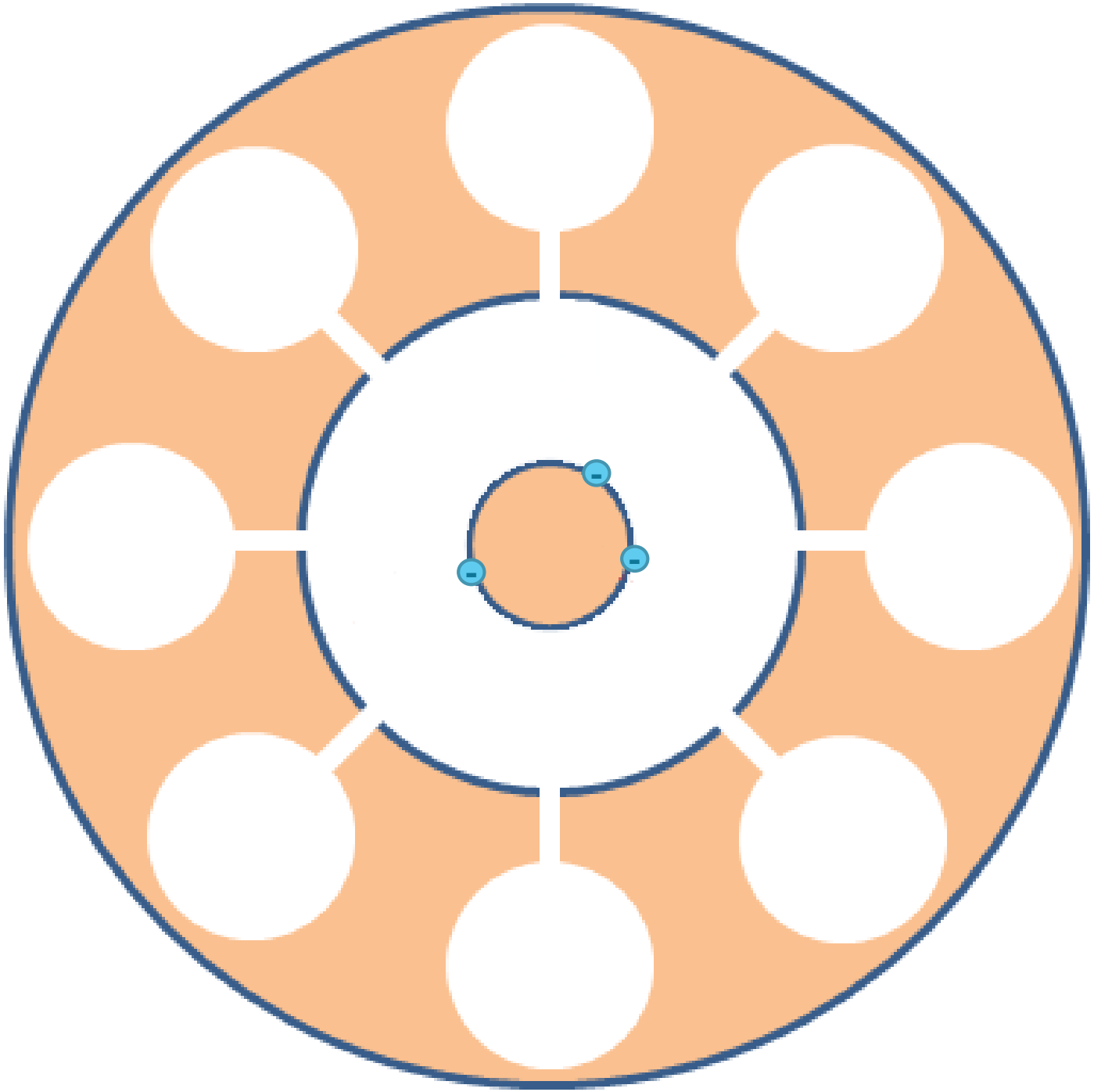
Magnetron - Components



- Negatively charged anode, inside cylindrical positively charged cathode
  - Cavities drilled into cathode
- Strong external magnetic field, along cylinder axis
- Strong electric field between cathode and anode

Magnetron - Operation

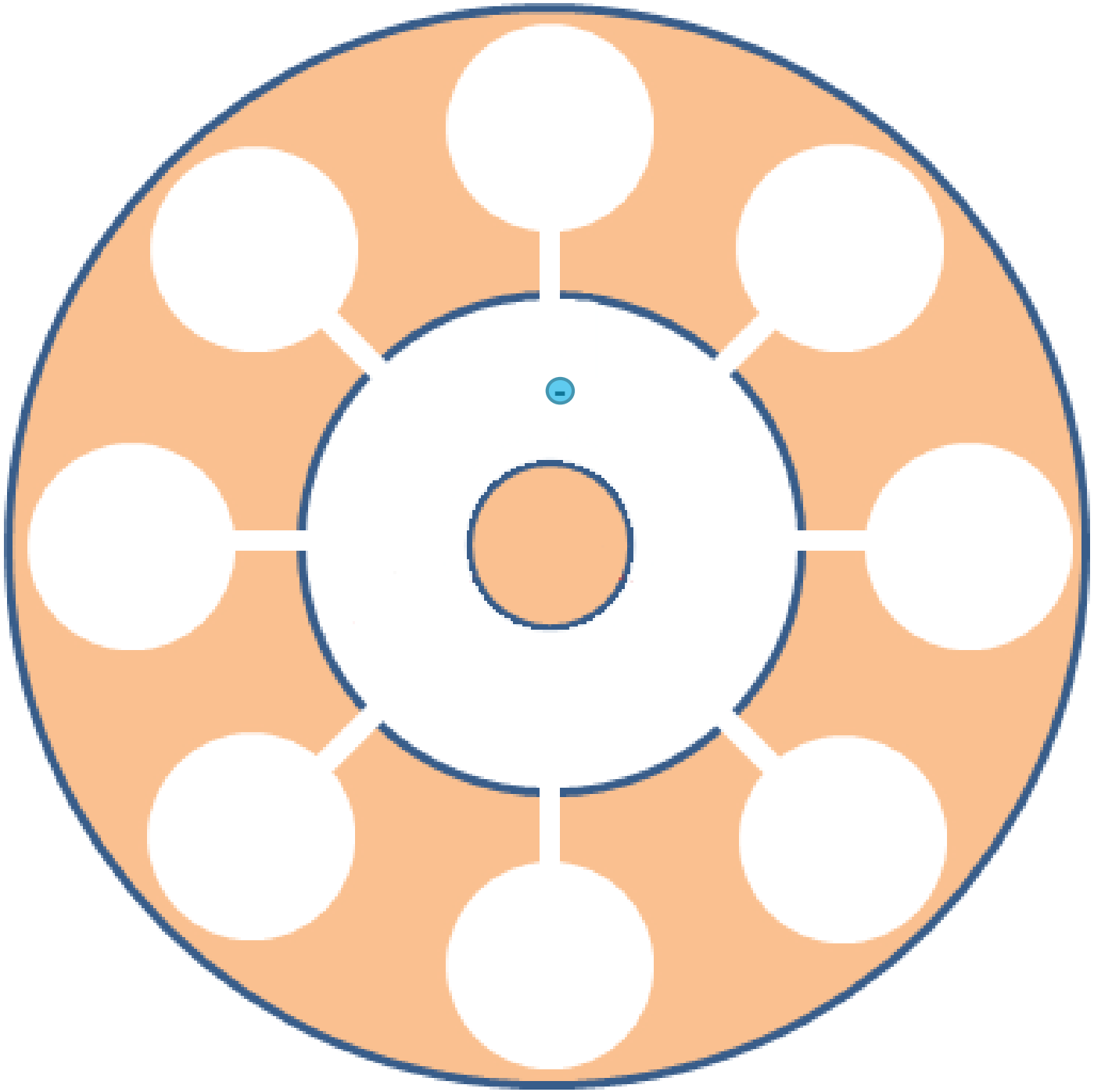
- Multiple interacting effects:
  - Electrons boil off anode



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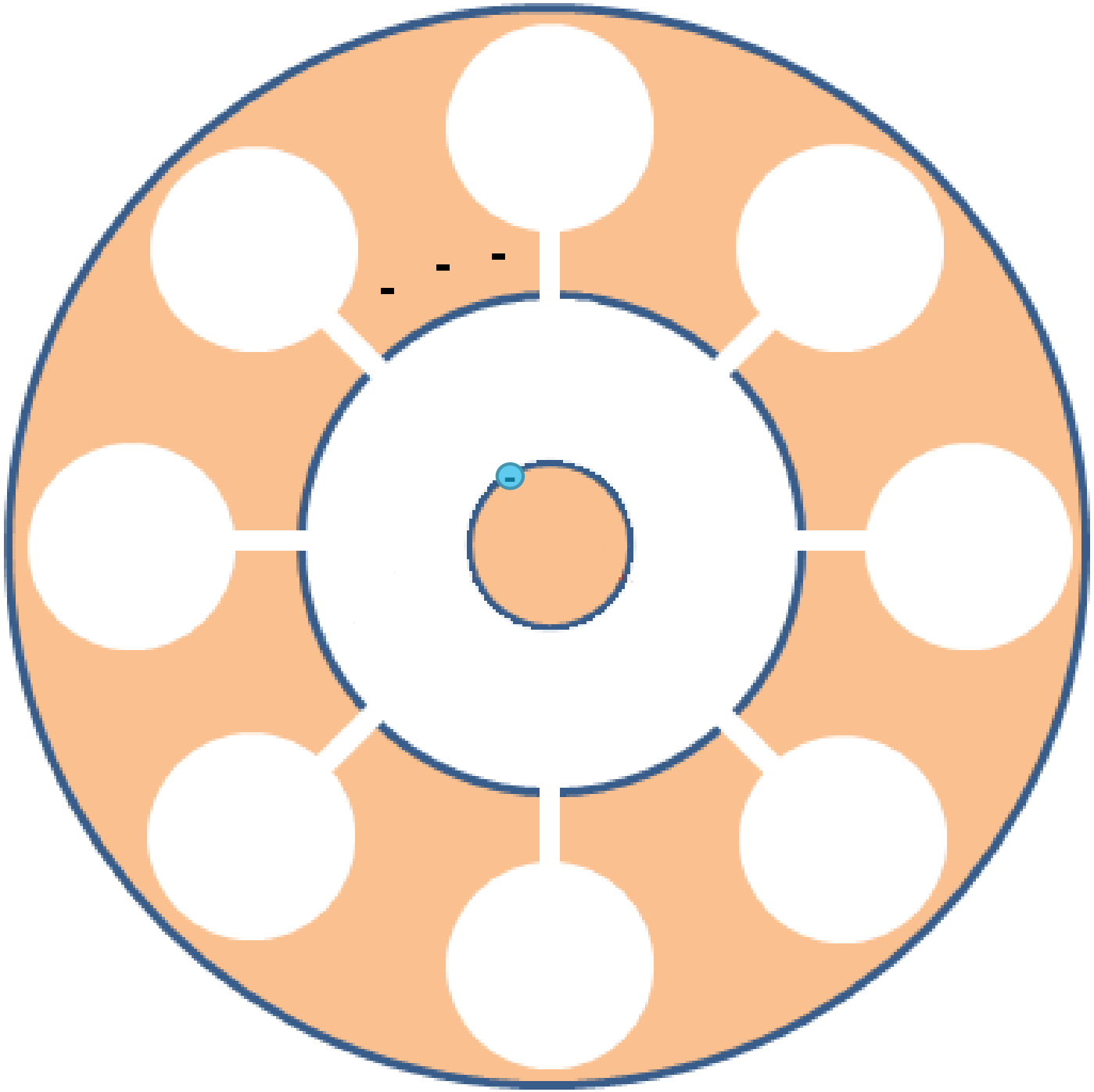
Magnetron - Operation

- Multiple interacting effects:
  - Electrons boil off anode
  - Drift in magnetic and electric fields



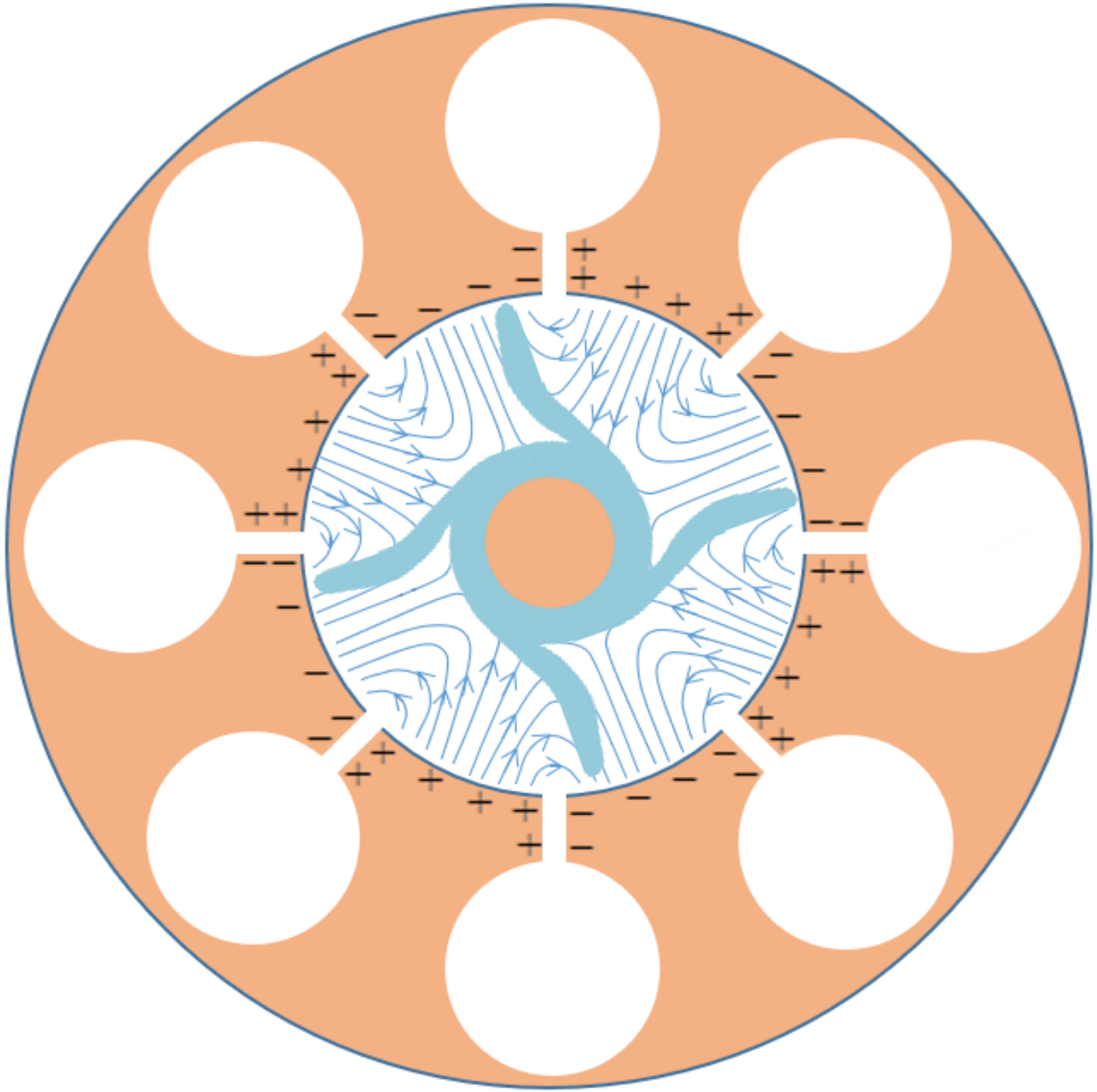
Magnetron - Operation

- Multiple interacting effects:
  - Electrons boil off anode
  - Drift in magnetic and electric fields
  - Electrons hit and charge the cavities



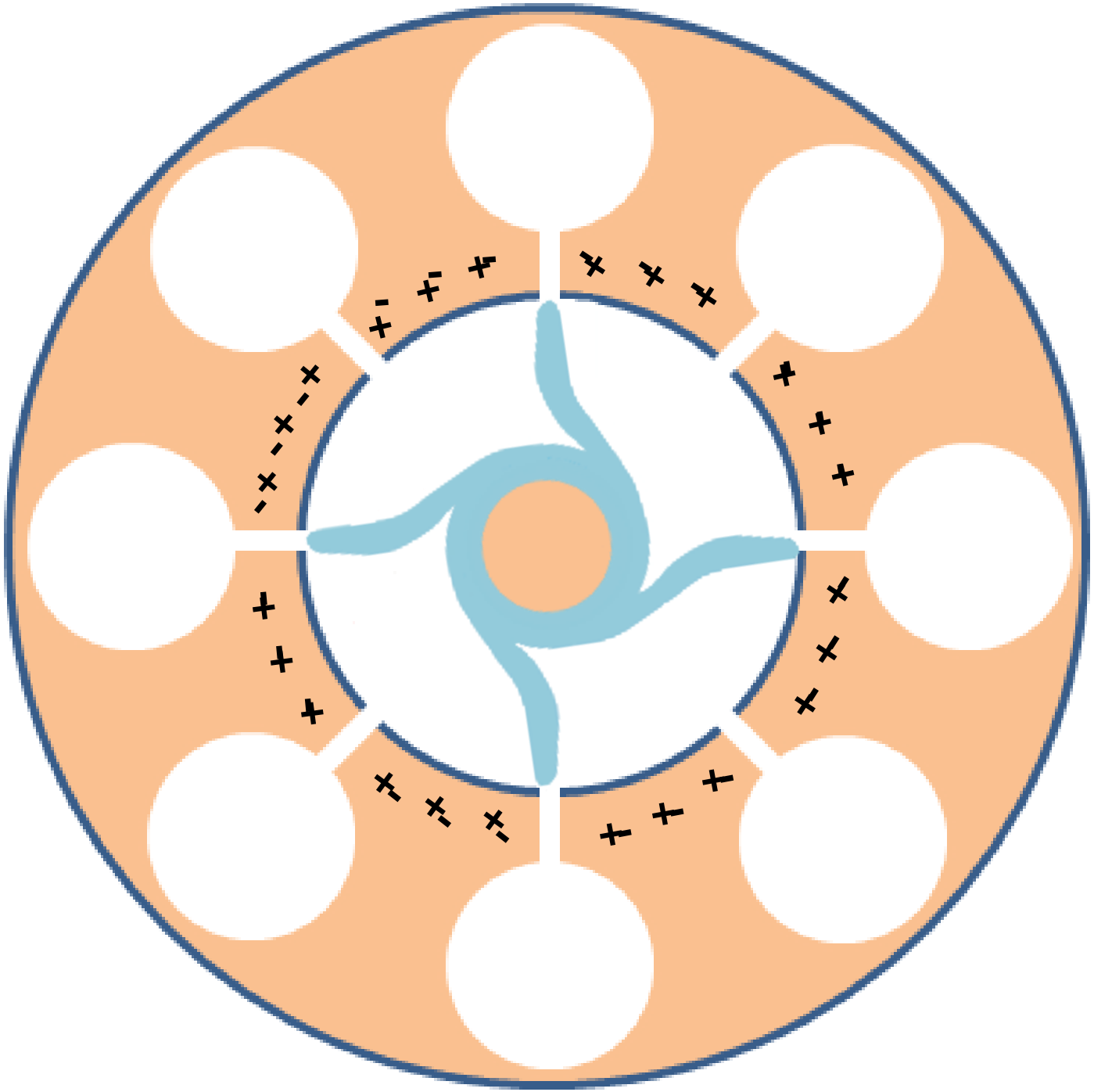
Magnetron - Operation

- Multiple interacting effects:
  - Electrons boil off anode
  - Drift in magnetic and electric fields
  - Electrons hit and charge the cavities
  - Cavities produce electric fields, focusing the electrons
    - Spontaneous symmetry breaking



Magnetron - Operation

- Multiple interacting effects:
  - Electrons boil off anode
  - Drift in magnetic and electric fields
  - Electrons hit and charge the cavities
  - Cavities produce electric fields, focusing the electrons
  - Cavities act as resonators, forcing periodic motion
  - External magnetic field rotates the ensemble



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Part B: Absorption of Radiation

Absorption of Microwave Radiation

- Common misconception:
  - Water heats up in microwave ovens due to absorption lines.
- Actually, due to dielectric absorption.
  - Not even the best frequency for absorption by water!
  - Water is partially transparent => more even cooking.

- Aim: Quantitative understanding of dielectric absorption of microwave radiation by water.
  - Dependence of absorption on temperature, salinity

Absorption of Microwave Radiation - Principles

- Water is treated as ensemble of interacting dipoles.
- Traveling EM wave rotates dipoles.
- Interaction between dipoles introduces delay in response.
- Phase difference produces heat.

Questions?



Thermoacoustic Engine

Theory Question 3

By: Pavel Radzivilovsky, Gal Dor,
Niv Cohen, Ron Lifshitz, Guy Segal



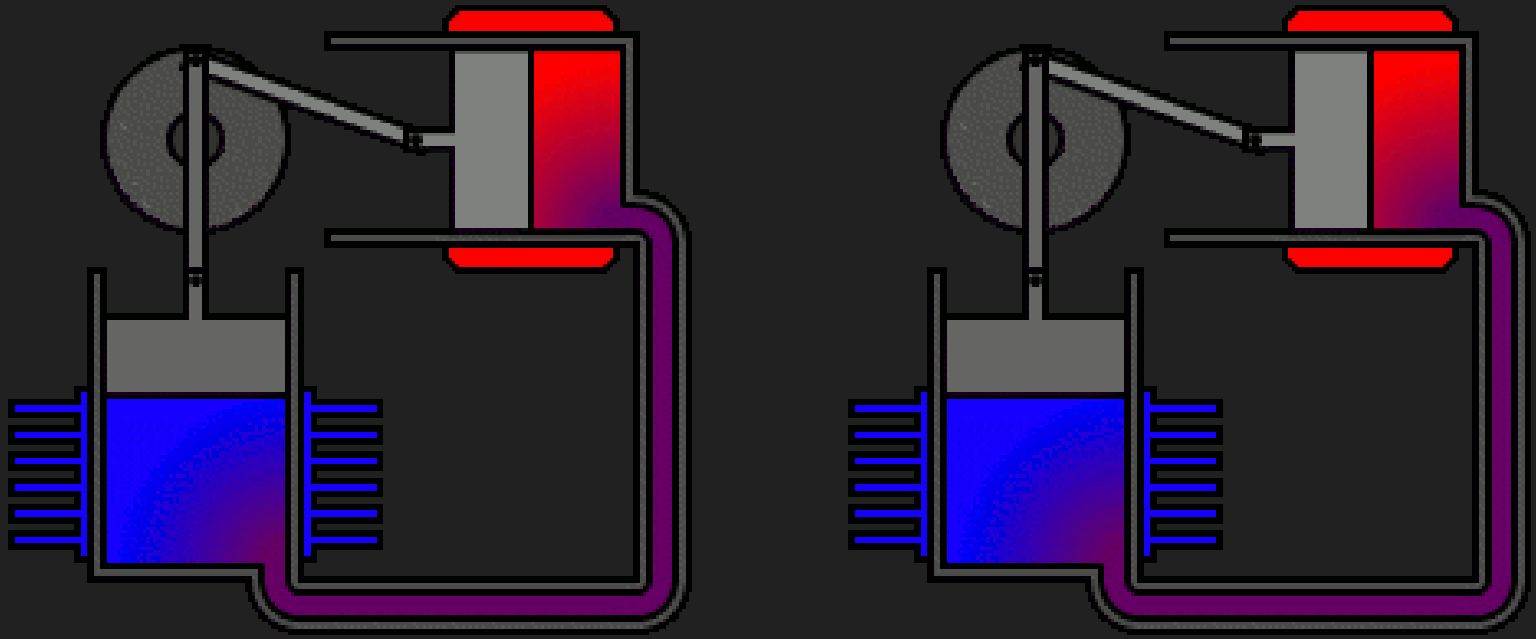
Solar-thermal



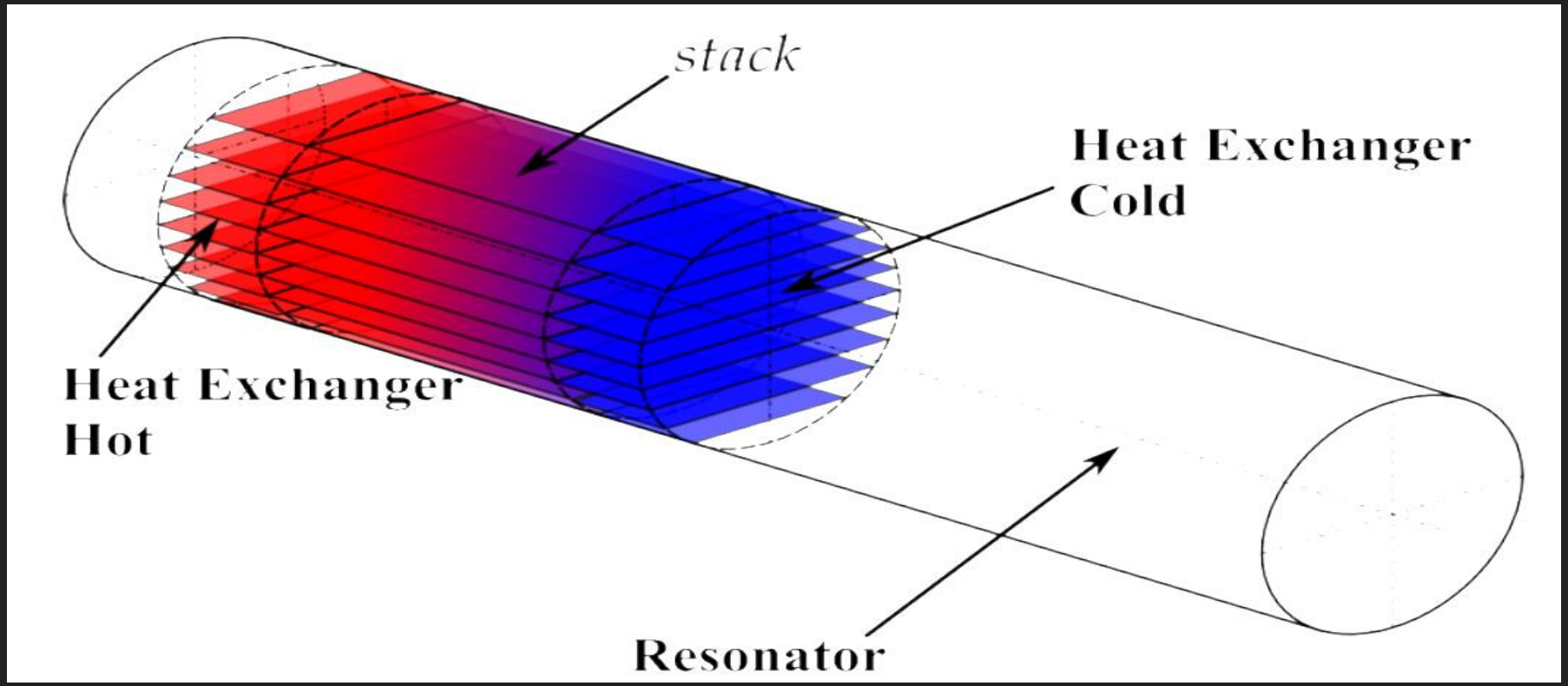
Waste heat



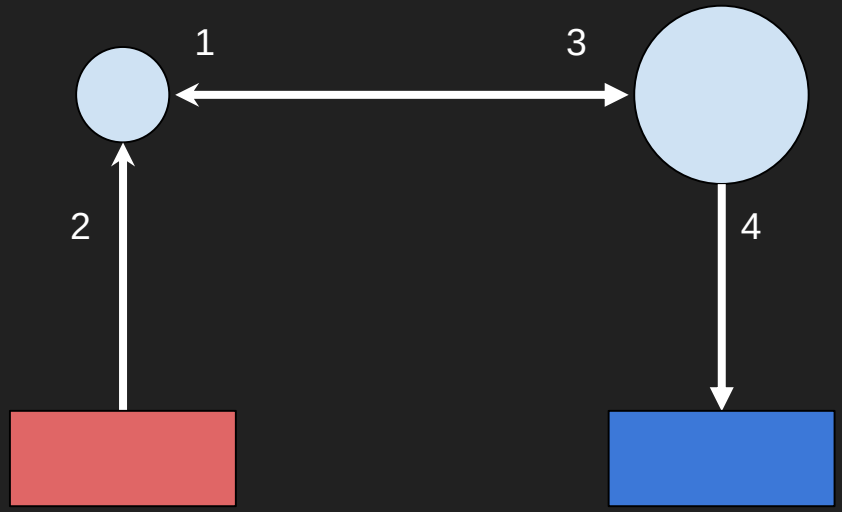
Inspiration: the Stirling engine



Standing Wave Thermoacoustic Device







The Thermoacoustic Cycle

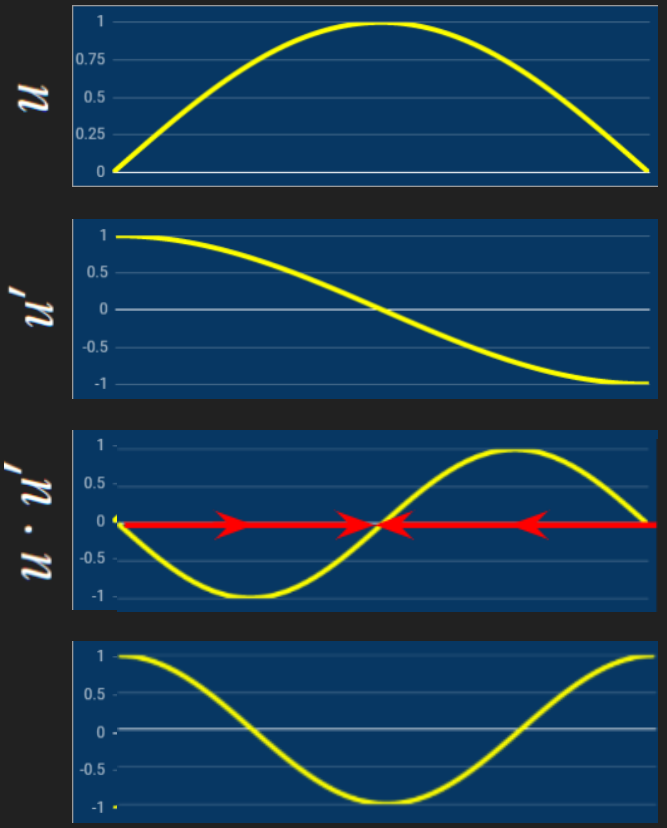
The stroke boundaries are “blurred”

1. Compression
2. Heat injection
3. Expansion
4. Heat removal

And now, let’s look at the question...

Please open the Theory Question 3 to the page 1..

Part A: Standing wave, no heat exchangers



A1. $u \propto sin(kx)$

A2. $\Delta V \propto u' \propto cos(kx)$

A3. $ma = F = -SP'\Delta x$

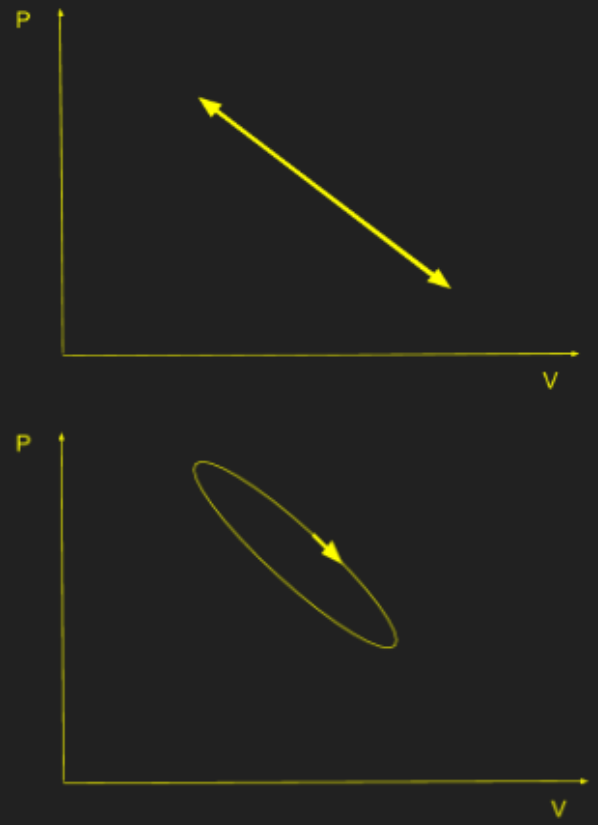
A4. $\Delta P/P = -\gamma \Delta V/V$

A5. $\Delta T \propto \Delta P \propto -\Delta V \propto u'$

A6. *Heat flux* $\propto u \cdot u'$

$$\sin(kx) \cdot \cos(kx) \propto \sin(2\,kx)$$

Part B: The self-amplifying mode of oscillations



B1-B2. $T_{st} > T_1$

B3. The 1st Law,

a.k.a. energy conservation

B4. Solve B3 for both phases

B5-B7. Integrate some cosines

to get W, Q, η